VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELAGAVI-590 014





A Project (phase-1) Synopsis Report

On

"Machine Learning–Based Dynamic Pricing Model for E-Commerce"

A project (phase-1) report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering in Computer Science & Engineering

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Project title

Machine Learning–Based Dynamic Pricing Model for E-Commerce

Problem Statement

With the rapid growth of online shopping platforms and digital marketplaces, pricing strategies have become more complex and competitive. E-commerce companies such as Amazon and Flipkart frequently adjust product prices based on demand, competitor pricing, customer behavior, seasonal trends, and stock availability. Traditional static pricing methods are no longer effective because they fail to respond quickly to market changes and customer preferences.

Many businesses still rely on manual or rule-based pricing systems, which may lead to overpricing (resulting in loss of customers) or underpricing (resulting in reduced profit margins). Additionally, sudden changes in demand during festivals, sales, or peak seasons make it difficult to maintain an optimal price using conventional methods. This creates a strong need for an intelligent, data-driven pricing system that can analyze historical data, identify patterns, and automatically predict the most suitable price.

Therefore, developing a Machine Learning–based Dynamic Pricing Model is essential to help businesses make informed pricing decisions, improve revenue generation, and remain competitive in the digital marketplace.

Objectives of the System

- To design a machine learning–based dynamic pricing model.
- To develop a system that predicts optimal product prices based on demand, stock availability, and competitor pricing.
- To implement data preprocessing and feature engineering techniques for improving model performance.
- To evaluate the performance of the model using metrics such as MAE, RMSE, and R^2 score.
- To compare different machine learning algorithms and select the best-performing model.

Scope of the Project

What is Included

- Collection of historical dataset related to product pricing, demand, stock levels, and competitor prices (public dataset or synthetic data).

- Data preprocessing techniques such as handling missing values, encoding categorical variables, normalization, and feature scaling.

- Exploratory Data Analysis (EDA) to understand patterns, correlations, and pricing trends.

- Feature engineering such as creating demand-to-stock ratio, seasonal indicators, and trend-based features.

- Implementation of supervised machine learning algorithms such as: Liner Regression, Decision tree, Random forest.

- Splitting dataset into training and testing sets for proper evaluation.

- Performance evaluation using metrics like MAE, RMSE, and R^2 Score.

- Selection of the best-performing model based on accuracy and error comparison.

- Optional development of a simple web interface to demonstrate price prediction functionality.

- Documentation of system design, methodology, and performance analysis.

What is Not Included

- Real-time integration with live e-commerce platforms.
- Real-time competitor price scraping from websites.
- Large-scale distributed systems or cloud-based production deployment.
- Complex reinforcement learning or deep learning–based advanced pricing strategies.
- Real-time payment gateway or inventory management system integration.
- Full-scale enterprise-level commercial implementation.

Platforms:

- Windows or Linux operating system
- Python programming environment
- Jupyter Notebook or VS Code for development
- Optional Streamlit platform for web-based interface

Tools and Technologies:

- **Python** – Main programming language for implementation.
- **Pandas** – Data manipulation and preprocessing.
- **NumPy** – Numerical and mathematical operations.
- **Matplotlib / Seaborn** – Data visualization and trend analysis.
- **Scikit-learn** – Machine learning model development and evaluation.
- **Streamlit (Optional)** – For building a simple interactive web interface.
- **CSV Dataset** – For storing and processing historical pricing data.

Assumptions:

- Historical sales data is available and reliable.
- Input features such as demand, stock, and competitor price significantly influence product pricing.
- Market conditions are relatively stable within the dataset period.
- The dataset is sufficient in size to train and evaluate the model effectively.
- The model predictions are used for recommendation purposes rather than final automated pricing in a real commercial system.

Methodology / Approach

The project will follow a structured multi-phase approach:

Phase 1: Problem Analysis and Planning.

- Identify the business problem of static pricing.
- Define the goal of the system (predict optimal price).
- Identify key factors affecting price such as demand, stock, competitor price, and seasonal trends.
- Plan the tools, dataset, and development approach.

Output of this phase: Clear understanding of problem and project plan.

Phase 2: Data Collection and Preparation

- Collect dataset from a public source (Kaggle) or create a synthetic dataset.
- Load data using Python (Pandas).
- Clean the data by:

  * Removing missing values

  *Handling incorrect entries

  *Converting categorical data into numeric format

- Perform basic visualization to understand patterns.

Phase 3: Feature Engineering and Analysis

- Analyze relationships between variables (e.g., demand vs price).
- Create new useful features such as:

  * Demand-to-stock ratio

  * Competitor price difference

  * Seasonal indicators

- Select important features that influence pricing.

Phase4: Model Development and Training

- Split the dataset into training and testing sets (80%–20%).
- Implement machine learning algorithms such as:

  *Linear Regression

  *Decision Tree

  *Random Forest
- Train the models using training data.

Phase5: Model Evaluation and Optimization

- Test models using testing data.
- Evaluate performance using:

  Mean Absolute Error (MAE)

  Root Mean Square Error (RMSE)

  R^2 Score
- Compare model performances.
- Select the best-performing model.

Phase 6: Implementation and Demonstartion

- Develop a simple interface (optional using Streamlit).
- Allow users to input demand, stock, and competitor price.
- Display predicted optimal price.
- Prepare documentation and final report.

Expected Outcomes / Deliverables

The expected outcomes of this project include:

- A functional dynamic pricing prediction system.
- A trained and evaluated machine learning model capable of predicting optimal product prices.
- Performance analysis report showing accuracy and error metrics.
- Optional: A simple web-based interface demonstrating price prediction functionality.
- A comprehensive project report documenting methodology, results, and analysis.

Relevance to the Field

This project is highly relevant to the field of Computer Science as it applies theoretical concepts of machine learning, data analysis, and software development to solve a real-world business problem. It involves database handling, data preprocessing, algorithm implementation, model evaluation, and system design.

The project enhances practical knowledge of AI/ML techniques, computing methodologies, and performance evaluation. It also demonstrates the application of programming skills and analytical thinking in building an intelligent decision-making system, which aligns with the learning outcomes of the course.

Keywords

- Dynamic Pricing
- Machine Learning
- Price Optimization
- Data Analysis
- Revenue Maximization
- Python

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